

8. (1.25 points)

a. (0.25 point)

Briefly explain when a curve-fitting method for selecting loss emergence patterns will produce a higher mean estimate of ultimate losses than a weighted average method.

b. (0.5 point)

Identify one reason why each of the methods in part a. above might be better than the other for estimating the payment pattern.

c. (0.5 point)

Briefly explain why the standard deviations of the ultimate losses for each of the scenarios below are narrower than the standard deviation of the ultimate loss for the loss development method using a curve fit to derive the emerged percentages:

- i. Clark Cape Cod method using a curve fit to derive the emerged percentages.
- ii. Loss development method using weighted averages of the development factors.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 8	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE(S): A2
SAMPLE ANSWERS	
Part a: 0.25 point	
<u>Sample Responses</u> - When weighted average LDF won't generate a tail factor (not enough data to model tail factor). But, curve-fitting will naturally generate a tail factor > 1, causing the mean to be higher than weighted average. - If there is a negative development due to salvage and subrogation. Curve fitting method assumes the % paid will always go up, but weighted average will recognize it (negative development). - If your weighted average method is weighting more recent years at a higher weight, and loss developments have dropped off drastically in recent years. - If there is an outlier age to age factor that is really low and has a lot of weight it will sway the weighted average method downward while the curve fit doesn't react to deviations as much.	
Part b: 0.5 point	
<u>Sample Responses</u> Weighted average method - Weighted average estimate relies upon actual data of company to produce estimates - Weighted average method is easy to understand and explain to people. - Weighted average makes more sense when there's negative development due to S&S since curve-fitting assumes increasing development. - Weighted average is easier to calculate, as you don't have to fit a curve at all. - If the payment patterns are relatively stable and sufficient data, then the weighted average method is likely better. - Weighted average is good when there is a change in the development pattern, since the weighted average allows us to put more weight on recent years. Curve fit method - Curve fit provides an estimate of development after the end of the available data. - Curve fit smooths out any spurious points in the data. - Curve fitting is better because it can work with any development age, while weighted average will only work with the same development age as the data. - Curve fitting: less parameters reduce the variance of the estimation (parameter variance). - Can be used on data in other than triangle format	
Part c: 0.5 point	
<u>Sample responses for part (i): vs. Clark Cape Cod method using a curve fit</u> - The cape cod method uses less parameters and that will reduce variability (and also standard deviation). - Cape Cod uses additional info, like exposure and on-level premium.	

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Sample responses for part (ii) vs. Loss development method using weighted averages of the development factors

- Using weighted averages of development factors produces a narrower range than using a curve fit because it ignores volatility in the tail.
- In this case, we don't have the additional uncertainty of estimating parameters for the development pattern curve, so parameter variance and therefore standard deviation of total losses, is reduced.

EXAMINER'S REPORT

Candidates were expected to know the pros and cons of using the two curve fitting methods described in the Clark paper and the impact of using such curves vs the simpler weighted average LDFs method.

Part a

Candidates were expected to give one example of when the curve fit method would produce a higher mean estimate of ultimate losses than the weighted average LDFs method.

Common mistakes included:

- Explaining that the weighted average LDFs method would respond to an accident year or a calendar year trend better than the curve fitting method. This is true only in certain circumstances, when we are putting more weight on relevant experience.

Part b

Candidates were expected to provide an example of situation where each of the curve fit method and the weighted average LDFs method is superior.

Common mistakes included:

- Only providing an example for one of the two methods.
- Explaining that the weighted average LDFs method would respond to an accident year or a calendar year trend better than the curve fitting method (similar to part a). This is true only in certain circumstances, when we are putting more weight on relevant experience.

Part c

Candidates were expected to explain why the standard deviation of the two methods provided was lower than the standard deviation of the LDF curve fit method.

Candidates had more difficulty responding for the LDF method with weighted average development patterns than the Cape Cod method with curve fit.

Common mistakes included:

- Stating that the weighted average method was the Minimum Variance Unbiased Linear Estimator as per the Mack paper. This is not a relevant reason because the question is not comparing two linear estimators (LDF with curve fit method is not a linear method).
- Explaining that the regular weighted average method relied more on actual data, thus reducing the SSE, but the LDF curve fit also relies on actual data to the same extent. Also, having a lower SSE does not necessarily lead to a lower variance: the Clark paper specifically states that overfitting the data usually leads to higher parameter variance, thus increasing overall variance.